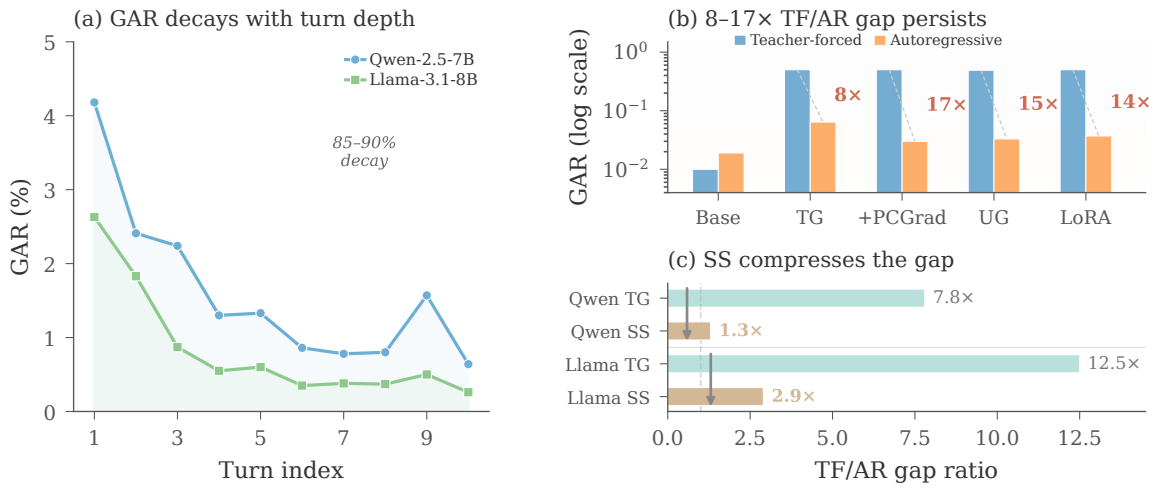
(a) GAR decays with turn depth
Qwen-2.5-7B
Llama-3.1-8B
85–90% decay
GAR (%)
Turn index
(b) 8–17× TF/AR gap persists
Teacher-forced
Autoregressive
GAR (log scale)
8×
17×
15×
14×
Base
TG
+PCGrad
UG
LoRA
(c) SS compresses the gap
Qwen TG
Qwen SS
Llama TG
Llama SS
7.8×
1.3×
12.5×
2.9×
TF/AR gap ratio